EXAMPLE 10.9

Consider two springs whose force constants are 1 N m^{-1} and 2 N m^{-1} which are connected in series. Calculate the effective spring constant (k_s) and comment on k_s.

Solution

$$k_1 = 1 \text{ N m}^{-1},\ k_2 = 2 \text{ N m}^{-1}$$

$$k_s = \frac{k_1 k_2}{k_1 + k_2} \text{ N m}^{-1}$$

$$k_s = \frac{1 \times 2}{1+2} = \frac{2}{3} \text{ N m}^{-1}$$

$$k_s < k_1 \text{ and } k_s < k_2$$

Therefore, the effective spring constant is lesser than both k_1 and k_2.

b. Springs connected in parallel

When two or more springs are connected in parallel, we can replace (by removing) all these springs with an equivalent spring (effective spring) whose net effect is same as if all the springs are in parallel connection. Given the values of individual spring constants to be k_1, k_2, k_3, ... (known quantities), we can establish a mathematical relationship to find out an effective (or equivalent) spring constant k_p (unknown quantity). For simplicity, let us consider only two springs of spring constants k_1 and k_2 attached to a mass m as shown in Figure 10.19. The results can be generalized to any number of springs in parallel.

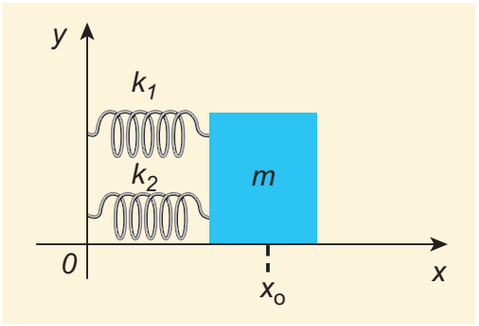


Figure 10.19 Springs connected in parallel

Let the force F be applied towards right as shown in Figure 10.20. In this case, both the springs elongate or compress by the same amount of displacement. Therefore, net force for the displacement of mass m is

$$F = -k_p x \qquad (10.45)$$

where k_p is called ***effective spring constant***. Let the first spring be elongated by a displacement x due to force F_1 and second spring be elongated by the same displacement x due to force F_2, then the net force

$$F = -k_1 x - k_2 x \qquad (10.46)$$

Equating equations (10.46) and (10.45), we get

$$k_p = k_1 + k_2 \qquad (10.47)$$

Generalizing, for n springs connected in parallel,

$$k_p = \sum_{i=1}^{n} k_i \qquad (10.48)$$

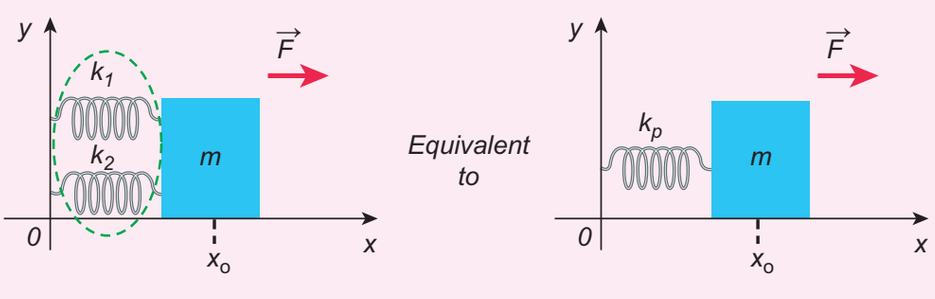


Figure 10.20 Effective spring constant in parallel connection

UNIT-10(XI-Physics_Vol-2).indd 205 25-03-2022 20:30:14